1 I removed the lid and handles from a domestic pressure cooker and used the cooker's body as the basis for my chamber. Do not use ordinary cooking pots as they are not made of heavy enough material to withstand the vacuum and may deform or implode. While this is unlikely to cause any danger, it will not result in a working chamber so don't waste time going down this route. Don't ask how I know this!

2 The fittings for connection and control of the vacuum supply are all ¼in BSP components readily available online or from specialist vacuum, plumbing or hydraulics suppliers. They consist of a vacuum gauge, two valves, a four-way connector, a straight connector and an adaptor to fit the inlet hose from the vacuum supply

3 The lid of the pressure cooker must be replaced with a sheet of thick, clear Perspex or similar clear acrylic material so that we can see what is happening inside the chamber. I recommend a minimum thickness of 20mm. I tried some 12mm material at first and, while this worked, it deflected rather too much under vacuum for my liking so I upgraded to 20mm. A local sign maker or plastics supplier may be able to provide you with an inexpensive offcut. I could have left it square but opted to turn it down to a more attractive circular shape. After finding the centre, mark out the shape with a marker pen taped to a pair of dividers

4 Using a 3.5mm drill bit in a pillar drill, bore a hole in the centre of the workpiece so that it can be attached to a jig on the bandsaw to be cut to shape. A narrow blade is needed in the saw to follow the curve without binding

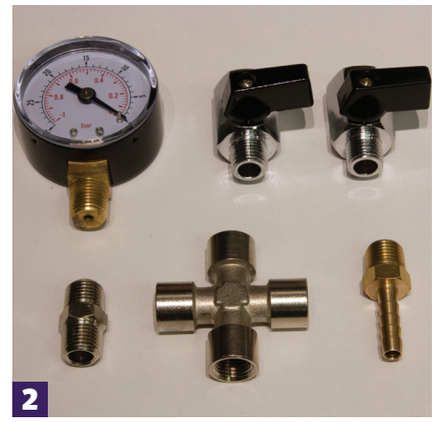
5 You could follow the line freehand, but setting up a jig ensures much greater accuracy. This simple arrangement consisted of a sheet of plywood clamped to the bandsaw table to which I was then able to screw the workpiece using a 3.5mm woodscrew through the hole pre-drilled in the centre. Careful alignment with the blade is required

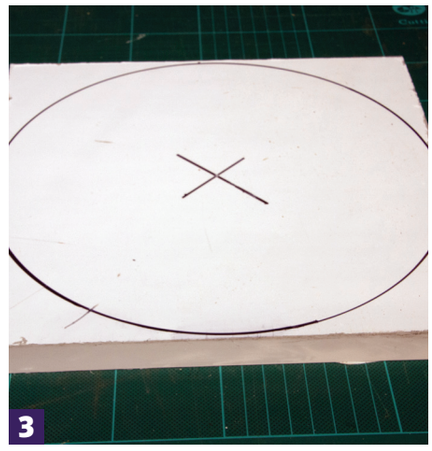
6 With the workpiece secured to the jig, turn on the saw and slowly turn the work into the blade to cut a perfect circle. Take care to keep your fingers well clear, turning the work by holding it at the side furthest from the blade

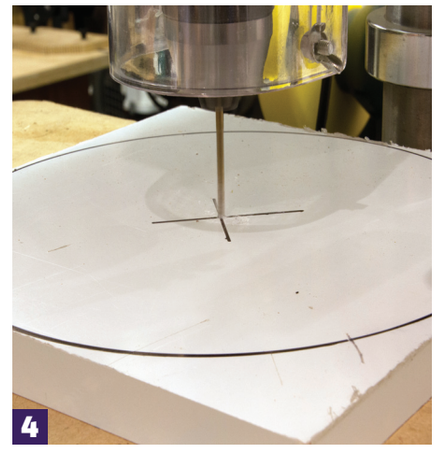
7 After completing the cut and removing the work from the saw, mount the Perspex on the lathe using the 3.5mm woodscrew as a mounting mandrel in a set of engineer's jaws fitted to a scroll chuck and supported by a revolving cone centre in the tailstock. You can then tidy up the sawn edges with the wing of a bowl gouge or whichever tool you prefer

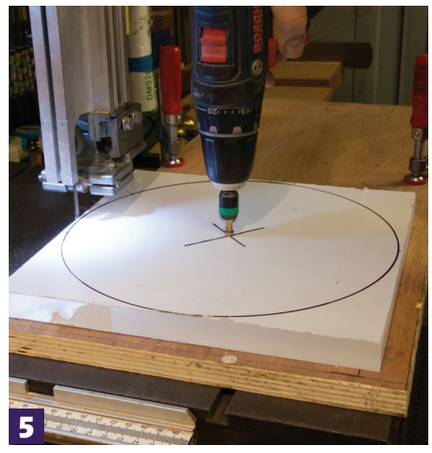
8 Sand the edge through the grits from 240 through to 600. I used Abranet, but any

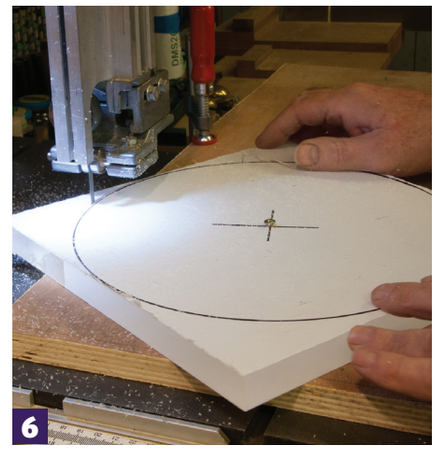
1


2


3


4


5


6

7

8

WT_289_21_24_PENS_TECH_WH_JRbdwhmb.indd 22 22/12/2015 09:56